

Investment Financing Conservatism and the Cross Section of Stock Returns

I. M. Harking

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Abstract

This paper studies the asset pricing implications of Investment Financing Conservatism (IFC), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023\)](#). A value-weighted long/short trading strategy based on IFC achieves an annualized gross (net) Sharpe ratio of 0.59 (0.47), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 22 (17) bps/month with a t-statistic of 3.18 (2.57), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Change in financial liabilities, Net external financing, Net debt financing, Asset growth, change in net operating assets, Employment growth) is 20 bps/month with a t-statistic of 2.90.

1 Introduction

Financial markets efficiently incorporate information about firm fundamentals into asset prices, yet the mechanisms through which production decisions affect expected returns remain central to understanding cross-sectional variation in equity returns. The production-based asset pricing literature demonstrates that firms’ investment and financing choices reflect their exposure to systematic risks, with these decisions serving as observable proxies for unobservable productivity shocks and adjustment costs that drive variation in discount rates (Cochrane, 1991; Zhang, 2005). Despite extensive evidence linking investment patterns to expected returns, researchers have devoted less attention to how firms’ conservative financing of capital expenditures—specifically their reluctance to fund investment through debt issuance—relates to their risk characteristics and subsequent stock returns.

Investment Financing Conservatism (IFC) captures firms’ propensity to fund capital expenditures through internal resources rather than external debt financing, reflecting fundamental differences in production technologies and exposure to aggregate productivity shocks. In production-based models, firms facing higher adjustment costs or greater uncertainty about future productivity optimally maintain financial flexibility by avoiding debt-financed investment, as external financing frictions amplify the real costs of capital adjustment (Gomes and Schmid, 2010; Livdan et al., 2009). These conservative firms exhibit lower operating leverage and reduced sensitivity to aggregate demand shocks, leading to lower systematic risk and expected returns compared to firms that aggressively lever their investment programs.

The cross-sectional variation in IFC directly maps to differences in firms’ marginal costs of production and their exposure to investment-specific technology shocks. Firms with high IFC ratios face binding financial constraints that prevent them from exploiting positive net present value projects, creating a wedge between their marginal product of capital and the risk-free rate (Whited and Wu, 2006; Hennessy

and Whited, 2007). This wedge varies systematically with aggregate conditions, as conservative firms' inability to time investment optimally exposes them to procyclical variation in the stochastic discount factor, generating time-varying risk premia that manifest in lower average returns.

The production-based interpretation of IFC connects directly to conditional risk premia through firms' optimal investment policies under uncertainty. When firms maintain conservative financing policies, they preserve the option value of waiting to invest until productivity shocks resolve, effectively holding real options that provide insurance against adverse aggregate shocks (Carlson et al., 2004; Cooper et al., 2008). This optionality reduces firms' systematic risk by allowing them to scale back investment during downturns without incurring financial distress costs, leading to lower loadings on production-based risk factors and consequently lower expected returns in equilibrium.

The Investment Financing Conservatism strategy generates economically and statistically significant returns across multiple specifications and risk adjustments. The value-weighted long/short portfolio that buys high IFC stocks and sells low IFC stocks earns an average monthly return of 29 basis points with a t-statistic of 4.26, translating to an annualized Sharpe ratio of 0.59. The strategy's performance remains robust after controlling for standard risk factors, with the monthly alpha relative to the Fama-French six-factor model equaling 22 basis points with a t-statistic of 3.18, confirming that IFC captures a distinct source of systematic risk not spanned by existing factors.

The economic significance of the IFC effect persists across different portfolio construction methodologies and size segments, addressing concerns about small-stock effects driving the results. Among the largest stocks—those with market capitalization above the 80th NYSE percentile—the IFC strategy achieves an average return of 39 basis points per month with a t-statistic of 4.43, while the six-factor alpha equals

31 basis points with a t-statistic of 3.49. These large-cap results demonstrate that the IFC premium reflects fundamental differences in firms’ production technologies rather than market microstructure effects or limits to arbitrage that primarily affect small stocks.

After accounting for transaction costs using the composite bid-ask spread measure, the net Sharpe ratio of 0.47 places the IFC strategy in the top 2% of documented anomalies, with the strategy maintaining positive and significant net alphas across all factor model specifications. The strategy’s alpha of 20 basis points per month (t-statistic = 2.90) remains significant even after controlling for the six Fama-French factors plus the six most closely related anomalies including change in net operating assets, net external financing, and asset growth. This orthogonality to related investment and financing variables confirms that IFC captures a unique aspect of firms’ production decisions that independently predicts cross-sectional returns.

This paper extends the production-based asset pricing literature by identifying Investment Financing Conservatism as a novel predictor of cross-sectional returns that captures firms’ joint investment and financing decisions within a unified production framework. While [Cooper et al. \(2008\)](#) document that asset growth predicts returns through its correlation with expected profitability, and [Lyandres and Palazzo \(2016\)](#) show that cash-financed asset expansions differ from debt-financed growth, our IFC measure explicitly links the financing source of capital expenditures to firms’ production technologies and conditional risk exposures. Unlike [Xing \(2008\)](#), who focus on the level of investment relative to assets, IFC emphasizes the interaction between real investment decisions and financial policy choices that jointly determine firms’ operating leverage and systematic risk.

Our methodological innovation lies in constructing a measure that simultaneously captures investment intensity and financing conservatism, providing a more complete characterization of firms’ production decisions than existing univariate sorting vari-

ables. The IFC signal’s robust performance after controlling for related anomalies from [Pontiff and Woodgate \(2008\)](#) on share issuance and [Hirshleifer et al. \(2012\)](#) on net operating assets demonstrates that the joint modeling of investment and financing decisions contains incremental information about expected returns. This finding supports theoretical predictions from dynamic models where financial frictions interact with real investment frictions to generate cross-sectional variation in risk premia.

The documented IFC premium has broader implications for understanding the sources of systematic risk in production economies and the channels through which corporate policies affect asset prices. By showing that conservative financing of investment predicts lower returns even among large, liquid stocks, we provide evidence that production-based factors operate through firms’ endogenous responses to productivity uncertainty rather than through financial market imperfections alone. These results suggest that incorporating firms’ dynamic financing choices into production-based asset pricing models could improve their ability to explain cross-sectional variation in expected returns, particularly for understanding how financial flexibility affects firms’ exposures to aggregate productivity and investment-specific technology shocks.

2 Data

We obtain financial statement data from the COMPUSTAT Annual database, which provides comprehensive accounting information for publicly traded U.S. companies. Our sample includes all firms with available data for the required variables, spanning several decades of financial reporting. We focus on non-financial firms and exclude observations with missing or negative values for the key components of our signal construction. Investment Financing Conservatism signal relies on two primary

variables from COMPUSTAT. Long-term debt issuance (DLTIS) represents the cash proceeds from the issuance of long-term debt during the fiscal year, capturing new borrowing activities that extend the firm’s long-term debt obligations. Capital expenditure (CAPX) measures the funds used by a company to acquire, upgrade, or maintain physical assets such as property, plants, buildings, technology, or equipment, representing the firm’s investment in productive capacity during the fiscal year. Both variables are reported in millions of dollars and reflect actual cash flows rather than accounting accruals. We construct the Investment Financing Conservatism signal by calculating the year-over-year change in long-term debt issuance scaled by lagged capital expenditure: $(DLTIS(t) - DLTIS(t-1)) / CAPX(t-1)$. This ratio captures the relationship between changes in external debt financing and the firm’s recent investment intensity. A higher value indicates that a firm is accelerating its long-term debt issuance relative to its capital investment base, potentially signaling more aggressive financing behavior, while a lower or negative value suggests conservative financing practices where debt issuance growth lags behind or decreases relative to capital investment levels. We compute this measure annually using fiscal year-end values and require non-missing observations for DLTIS in consecutive years and non-zero CAPX in the lagged period to ensure meaningful scaling. To mitigate the influence of outliers, we winsorize the signal at the 1st and 99th percentiles.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the IFC signal. Panel A plots the time-series of the mean, median, and interquartile range for IFC. On average, the cross-sectional mean (median) IFC is -3.65 (-0.00) over the 1973 to 2024 sample, where the starting date is determined by the availability of the input IFC data. The signal’s interquartile range spans -3.29 to 4.32. Panel B of Figure 1 plots the time-series of

the coverage of the IFC signal for the CRSP universe. On average, the IFC signal is available for 6.07% of CRSP names, which on average make up 7.19% of total market capitalization.

4 Does IFC predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on IFC using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high IFC portfolio and sells the low IFC portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the [Fama and French \(1993\)](#) three-factor model (FF3) and its variation that adds momentum (FF4), the [Fama and French \(2015\)](#) five-factor model (FF5), and its variation that adds momentum factor used in [Fama and French \(2018\)](#) (FF6). The table shows that the long/short IFC strategy earns an average return of 0.29% per month with a t-statistic of 4.26. The annualized Sharpe ratio of the strategy is 0.59. The alphas range from 0.22% to 0.34% per month and have t-statistics exceeding 3.18 everywhere. The lowest alpha is with respect to the FF6 factor model.

Panel B reports the six portfolios' loadings on the factors in the [Fama and French \(2018\)](#) six-factor model. The long/short strategy's most significant loading is 0.29, with a t-statistic of 6.28 on the CMA factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 504 stocks and an average market capitalization of at least \$1,502 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies

are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor’s achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using NYSE breakpoints and equal-weighted portfolios, and equals 19 bps/month with a t-statistics of 4.18. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-five exceed two, and for twenty-three exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns reported in the first column range between -8-25bps/month. The lowest return, (-8 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of -1.37. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the IFC trading strategy improves the achievable mean-variance efficient frontier spanned by the fac-

tor models in twenty cases, and significantly expands the achievable frontier in twenty cases.

Table 3 provides direct tests for the role size plays in the IFC strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and IFC, as well as average returns and alphas for long/short trading IFC strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the IFC strategy achieves an average return of 39 bps/month with a t-statistic of 4.43. Among these large cap stocks, the alphas for the IFC strategy relative to the five most common factor models range from 31 to 45 bps/month with t-statistics between 3.49 and 5.02.

5 How does IFC perform relative to the zoo?

Figure 2 puts the performance of IFC in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.¹ The vertical red line shows where the Sharpe ratio for the IFC strategy falls in the distribution. The IFC strategy’s gross (net) Sharpe ratio of 0.59 (0.47) is greater than 95% (98%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the IFC strategy (red line).² Ignoring trading costs, a \$1 invested in the IFC strategy

¹The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

²The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in [Detzel et al. \(2022\)](#), that

would have yielded \$4.37 which ranks the IFC strategy in the top 3% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the IFC strategy would have yielded \$2.73 which ranks the IFC strategy in the top 3% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and Novy-Marx and Velikov (2016) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the IFC relative to those. Panel A shows that the IFC strategy gross alphas fall between the 65 and 72 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 197306 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The IFC strategy has a positive net generalized alpha for five out of the five factor models. In these cases IFC ranks between the 79 and 87 percentiles in terms of how much it could have expanded the achievable investment frontier.

6 Does IFC add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations.

Figure 5 plots a name histogram of the correlations of IFC with 210 filtered anomaly

a capital cost is charged against the strategy's returns at the risk-free rate. This excess return corresponds more closely to the strategy's economic profitability.

signals.³ Figure 6 also shows an agglomerative hierarchical cluster plot using Ward’s minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price IFC or at least to weaken the power IFC has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of IFC conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{IFC} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{IFC}IFC_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{IFC,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on IFC. Stocks are finally grouped into five IFC portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted IFC trading strategies conditioned on each of the 210 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on IFC and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the IFC signal in these Fama-MacBeth regressions

³When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

exceed 1.76, with the minimum t-statistic occurring when controlling for change in net operating assets. Controlling for all six closely related anomalies, the t-statistic on IFC is 0.71.

Similarly, Table 5 reports results from spanning tests that regress returns to the IFC strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the IFC strategy earns alphas that range from 19-22bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 2.86, which is achieved when controlling for change in net operating assets. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the IFC trading strategy achieves an alpha of 20bps/month with a t-statistic of 2.90.

7 Does IFC add relative to the whole zoo?

Finally, we can ask how much adding IFC to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). The combinations use either the 155 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 155 anomalies augmented with the IFC signal.⁴ We consider one different methods for combining signals.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 155-anomaly combination strategy grows to \$993.77, while \$1 investment in the combination strategy that

⁴We filter the 207 [Chen and Zimmermann \(2022\)](#) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which IFC is available.

includes IFC grows to \$1241.53.

8 Conclusion

This study provides robust evidence that Investment Financing Conservatism (IFC) represents a significant and economically meaningful predictor of cross-sectional stock returns. Our comprehensive analysis, following the rigorous protocol of Novy-Marx and Velikov (2023), demonstrates that IFC maintains substantial predictive power even after accounting for transaction costs and controlling for established risk factors and closely related anomalies.

The key findings reveal that a value-weighted long/short strategy based on IFC generates impressive risk-adjusted returns, with an annualized net Sharpe ratio of 0.47 and statistically significant alphas relative to both the Fama-French five-factor model plus momentum (17 bps/month, t -statistic = 2.57) and an extended model including six closely related financing and investment strategies (20 bps/month, t -statistic = 2.90). The persistence of significant alphas even after controlling for related factors such as changes in financial liabilities, net external financing, and asset growth suggests that IFC captures a distinct dimension of firm behavior that is incrementally informative for expected returns.

From a practical perspective, these results indicate that IFC can serve as a valuable signal for portfolio construction and risk management. The strategy’s ability to generate positive risk-adjusted returns net of transaction costs suggests potential implementability for institutional investors. Moreover, the economic magnitude of the returns—while not extraordinary—appears sufficiently large to warrant consideration in multi-factor investment frameworks.

Several limitations of this study merit acknowledgment. First, while we employ conservative transaction cost estimates, actual implementation costs may vary

across different market conditions and investor types. Second, our analysis focuses primarily on U.S. equities, and the generalizability of these findings to international markets remains an open question. Third, the economic mechanisms underlying the IFC anomaly warrant deeper investigation to distinguish between risk-based and behavioral explanations.

Future research should explore several promising avenues. First, examining the performance of IFC across different market regimes and economic cycles could provide insights into its conditional effectiveness. Second, investigating the interaction between IFC and firm characteristics such as size, profitability, and financial constraints may reveal important heterogeneity in the signal's predictive power. Third, extending the analysis to international markets would test the robustness and universality of the IFC effect. Finally, developing theoretical models that can explain why conservative investment financing predicts returns would enhance our understanding of this anomaly and its place within the broader asset pricing literature. Such theoretical foundations could also guide the development of refined measures that might capture the underlying economic phenomenon more precisely.

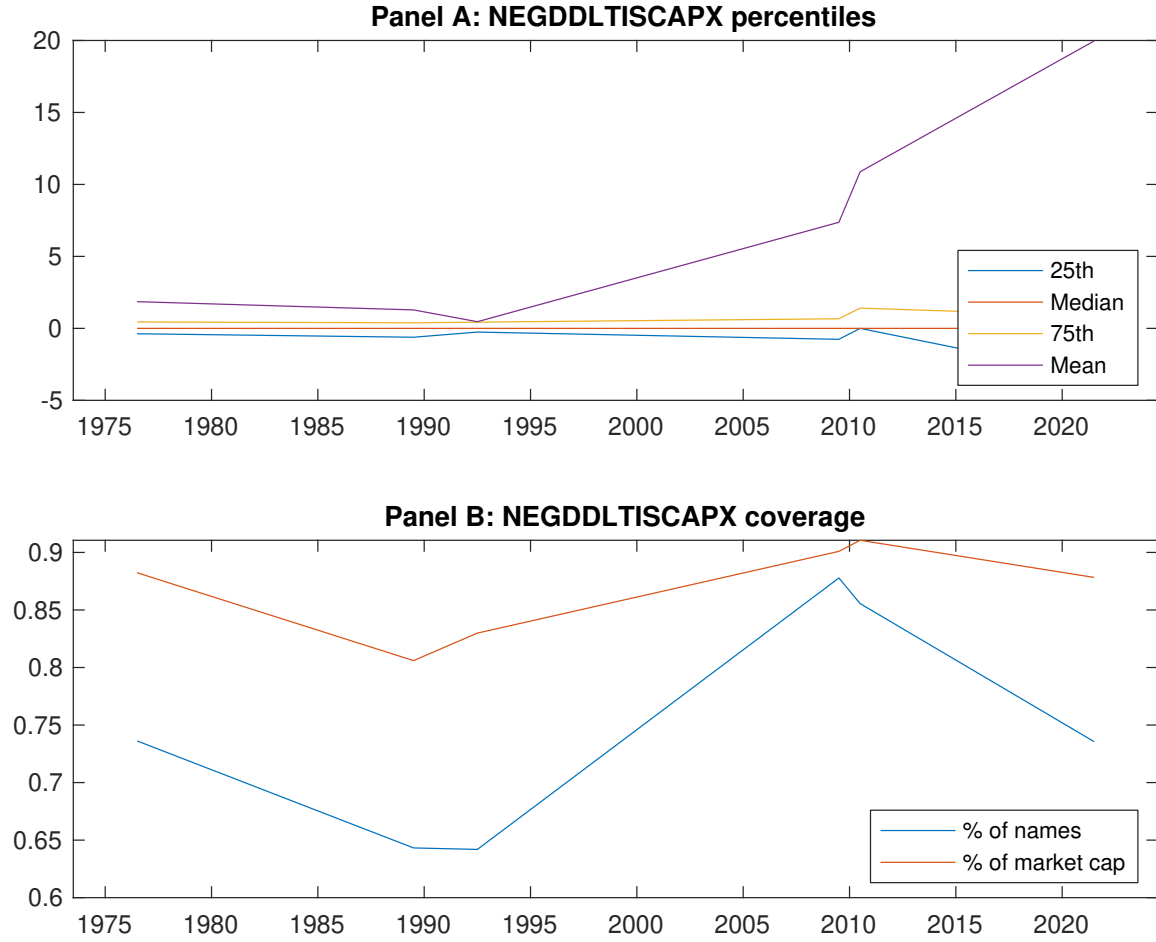


Figure 1: Times series of IFC percentiles and coverage. This figure plots descriptive statistics for IFC. Panel A shows cross-sectional percentiles of IFC over the sample. Panel B plots the monthly coverage of IFC relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on IFC. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, [Fama and French \(1993\)](#) three-factor model, [Fama and French \(1993\)](#) three-factor model augmented with the [Carhart \(1997\)](#) momentum factor, [Fama and French \(2015\)](#) five-factor model, and the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor following [Fama and French \(2018\)](#). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the [Fama and French \(2015\)](#) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 197306 to 202406.

Panel A: Excess returns and alphas on IFC-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.55 [2.56]	0.67 [3.76]	0.65 [3.26]	0.72 [4.05]	0.84 [4.19]	0.29 [4.26]
α_{CAPM}	-0.18 [-3.44]	0.07 [1.35]	-0.02 [-0.33]	0.12 [2.35]	0.16 [3.00]	0.34 [5.05]
α_{FF3}	-0.20 [-3.91]	0.04 [0.97]	0.05 [0.93]	0.12 [2.38]	0.13 [2.57]	0.34 [4.95]
α_{FF4}	-0.17 [-3.21]	0.04 [0.91]	0.10 [1.86]	0.07 [1.44]	0.13 [2.40]	0.30 [4.30]
α_{FF5}	-0.16 [-3.10]	-0.03 [-0.77]	0.10 [1.75]	0.03 [0.65]	0.08 [1.48]	0.24 [3.54]
α_{FF6}	-0.14 [-2.65]	-0.03 [-0.68]	0.13 [2.41]	0.00 [0.06]	0.08 [1.47]	0.22 [3.18]
Panel B: Fama and French (2018) 6-factor model loadings for IFC-sorted portfolios						
β_{MKT}	1.08 [89.35]	0.98 [95.69]	0.97 [75.06]	0.96 [83.22]	1.05 [84.68]	-0.03 [-1.99]
β_{SMB}	0.11 [6.08]	-0.12 [-7.63]	-0.02 [-0.83]	-0.02 [-1.12]	0.11 [6.04]	0.00 [0.09]
β_{HML}	0.08 [3.42]	0.06 [2.92]	-0.17 [-6.82]	-0.08 [-3.57]	-0.02 [-0.83]	-0.10 [-3.28]
β_{RMW}	0.01 [0.36]	0.14 [7.09]	-0.04 [-1.40]	0.08 [3.53]	0.08 [3.09]	0.07 [2.15]
β_{CMA}	-0.16 [-4.48]	0.10 [3.20]	-0.10 [-2.76]	0.22 [6.55]	0.13 [3.62]	0.29 [6.28]
β_{UMD}	-0.04 [-3.03]	-0.01 [-0.59]	-0.06 [-4.54]	0.05 [4.10]	-0.00 [-0.02]	0.04 [2.31]
Panel C: Average number of firms (n) and market capitalization (me)						
n	674	504	1021	562	636	
me (\$10 ⁶)	1502	2808	2233	2766	1568	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the IFC strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 197306 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.29 [4.26]	0.34 [5.05]	0.34 [4.95]	0.30 [4.30]	0.24 [3.54]	0.22 [3.18]
Quintile	NYSE	EW	0.19 [4.18]	0.21 [4.65]	0.20 [4.38]	0.19 [4.13]	0.18 [4.01]	0.18 [3.91]
Quintile	Name	VW	0.26 [3.58]	0.31 [4.26]	0.33 [4.46]	0.29 [3.88]	0.26 [3.63]	0.24 [3.31]
Quintile	Cap	VW	0.31 [4.74]	0.35 [5.46]	0.36 [5.68]	0.31 [4.82]	0.28 [4.41]	0.25 [3.87]
Decile	NYSE	VW	0.32 [3.64]	0.37 [4.16]	0.37 [4.21]	0.36 [4.04]	0.24 [2.78]	0.25 [2.85]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r_{net}^e	α_{CAPM}^*	α_{FF3}^*	α_{FF4}^*	α_{FF5}^*	α_{FF6}^*
Quintile	NYSE	VW	0.23 [3.36]	0.28 [4.07]	0.27 [3.93]	0.25 [3.61]	0.19 [2.81]	0.17 [2.57]
Quintile	NYSE	EW	-0.08 [-1.37]					
Quintile	Name	VW	0.20 [2.74]	0.24 [3.28]	0.25 [3.44]	0.23 [3.14]	0.21 [2.80]	0.19 [2.57]
Quintile	Cap	VW	0.25 [3.92]	0.29 [4.48]	0.30 [4.65]	0.27 [4.22]	0.23 [3.59]	0.20 [3.28]
Decile	NYSE	VW	0.25 [2.79]	0.29 [3.21]	0.29 [3.23]	0.29 [3.17]	0.18 [2.06]	0.18 [2.02]

Table 3: Conditional sort on size and IFC

This table presents results for conditional double sorts on size and IFC. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on IFC. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high IFC and short stocks with low IFC. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 197306 to 202406.

Panel A: portfolio average returns and time-series regression results												
Size quintiles	IFC Quintiles					IFC Strategies						
		(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
	(1)	0.72 [2.65]	0.90 [3.26]	0.91 [3.25]	0.95 [3.32]	0.77 [2.84]	0.05 [0.70]	0.07 [0.90]	0.05 [0.70]	0.05 [0.65]	0.08 [1.00]	0.08 [0.97]
	(2)	0.81 [3.06]	0.81 [3.20]	0.87 [3.41]	0.91 [3.64]	0.81 [3.24]	0.01 [0.08]	0.04 [0.52]	0.00 [0.04]	0.01 [0.16]	-0.04 [-0.50]	-0.03 [-0.36]
	(3)	0.74 [3.01]	0.85 [3.76]	0.80 [3.30]	0.79 [3.48]	0.93 [3.98]	0.19 [2.57]	0.23 [3.17]	0.22 [3.03]	0.18 [2.37]	0.21 [2.85]	0.18 [2.38]
	(4)	0.74 [3.34]	0.80 [3.76]	0.82 [3.71]	0.78 [3.63]	0.88 [4.07]	0.14 [1.88]	0.16 [2.19]	0.14 [1.93]	0.12 [1.68]	0.09 [1.26]	0.09 [1.15]
	(5)	0.45 [2.20]	0.63 [3.55]	0.59 [3.05]	0.62 [3.41]	0.84 [4.34]	0.39 [4.43]	0.42 [4.78]	0.45 [5.02]	0.37 [4.21]	0.36 [4.01]	0.31 [3.49]
Panel B: Portfolio average number of firms and market capitalization												
Size quintiles	IFC Quintiles					IFC Quintiles						
		Average n					Average market capitalization (\$10 ⁶)					
		(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)	
	(1)	380	381	381	381	377	35	34	31	33	34	
	(2)	105	105	105	105	105	60	60	59	60	59	
	(3)	75	75	75	75	75	106	106	103	104	106	
	(4)	63	63	63	63	62	226	234	225	230	224	
(5)	57	57	57	57	57	1328	2082	1748	2145	1446		

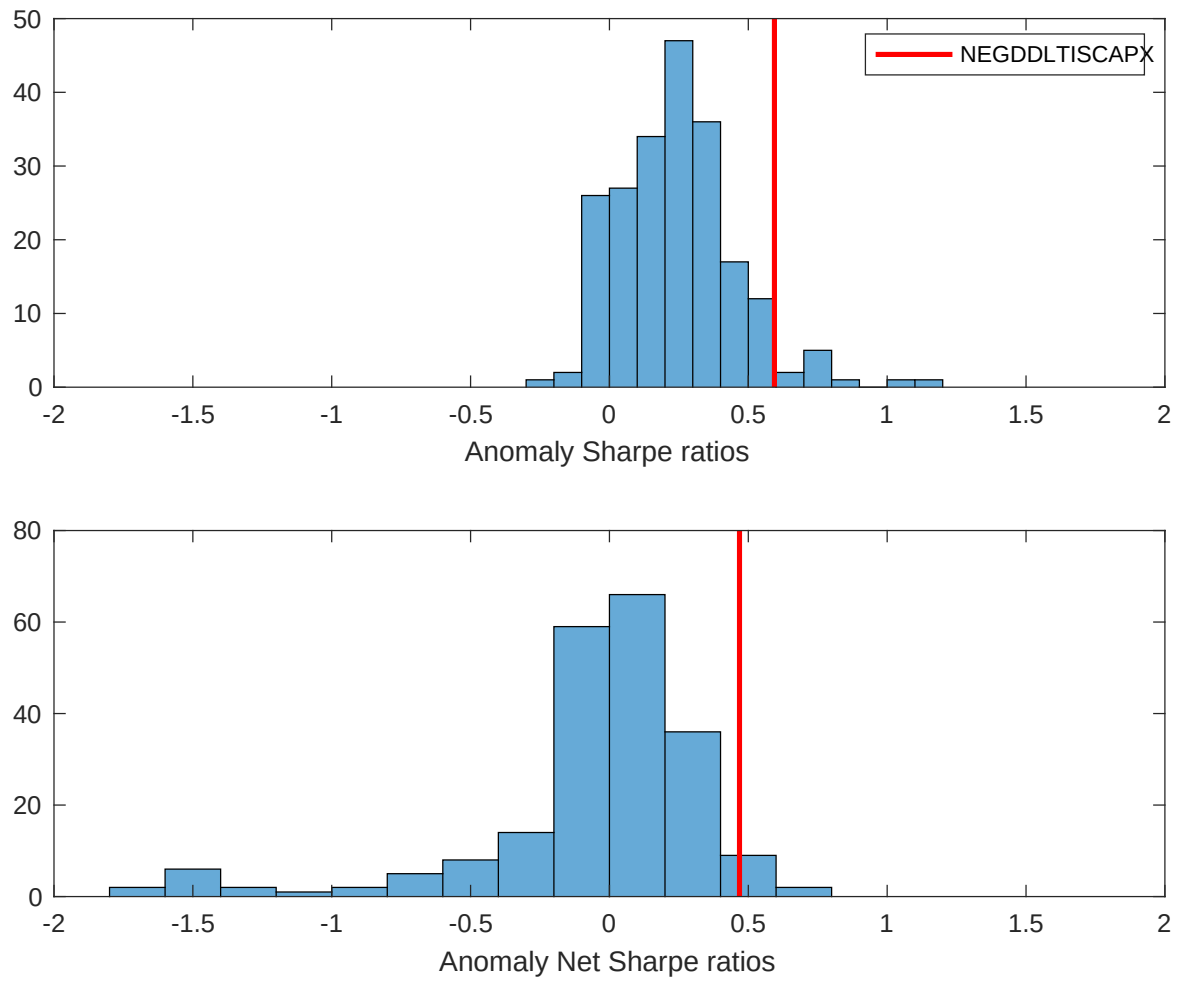


Figure 2: Distribution of Sharpe ratios.

This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the IFC with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

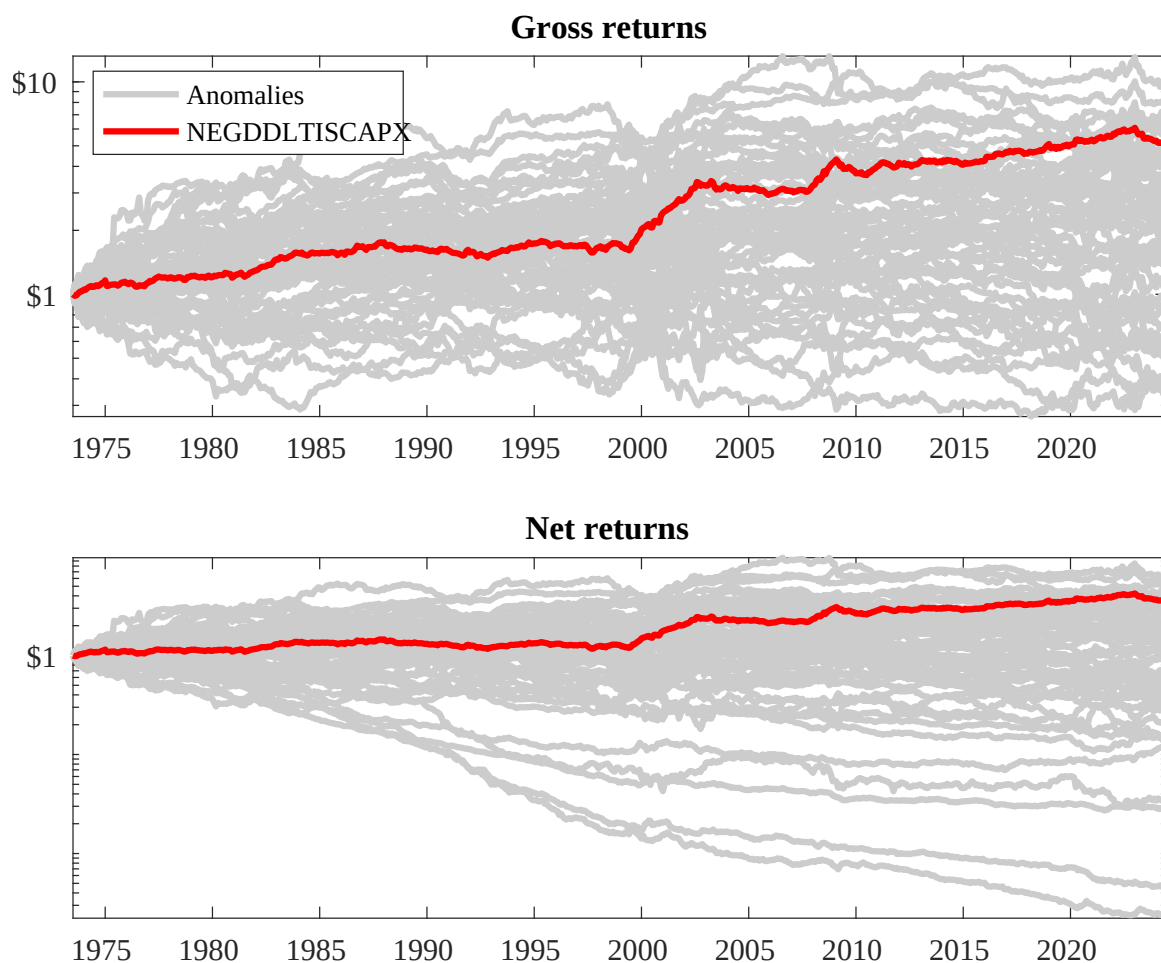
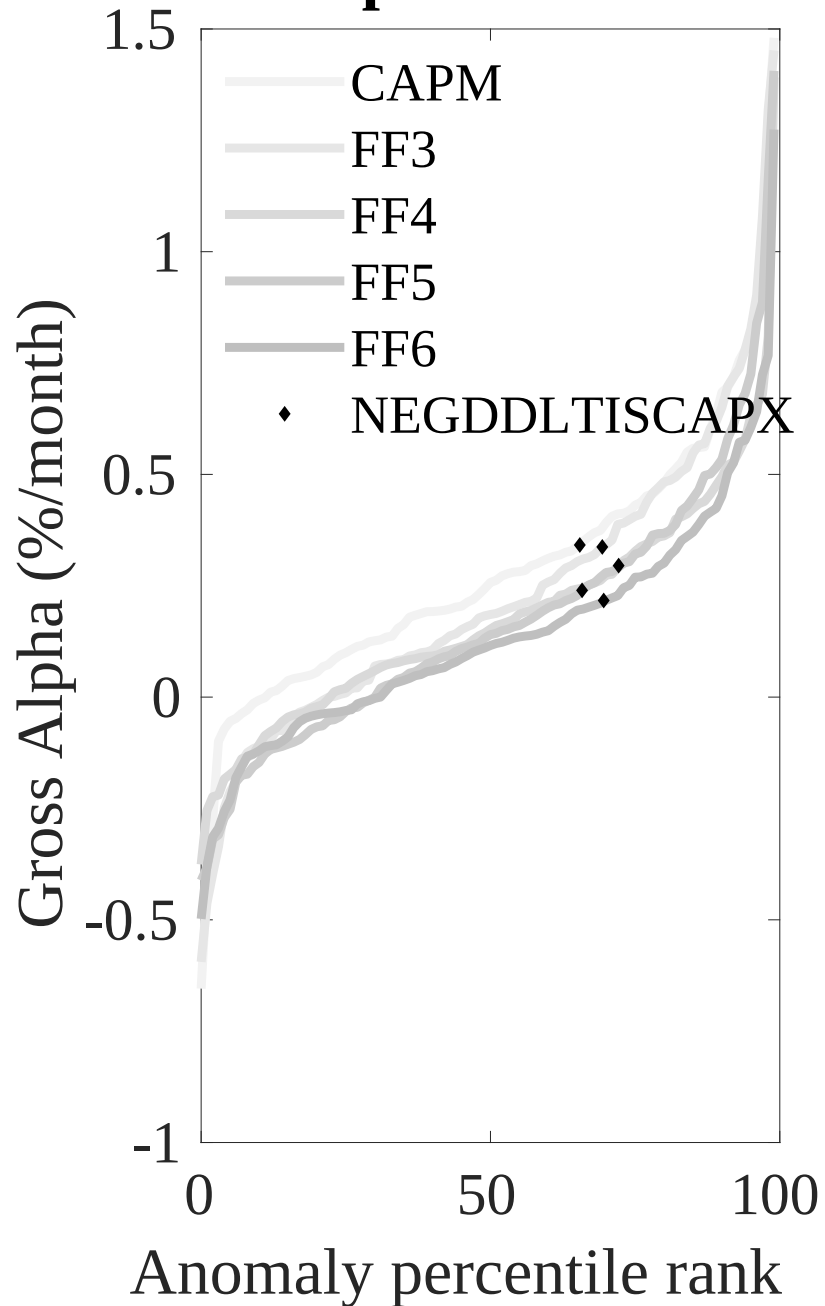


Figure 3: Dollar invested.

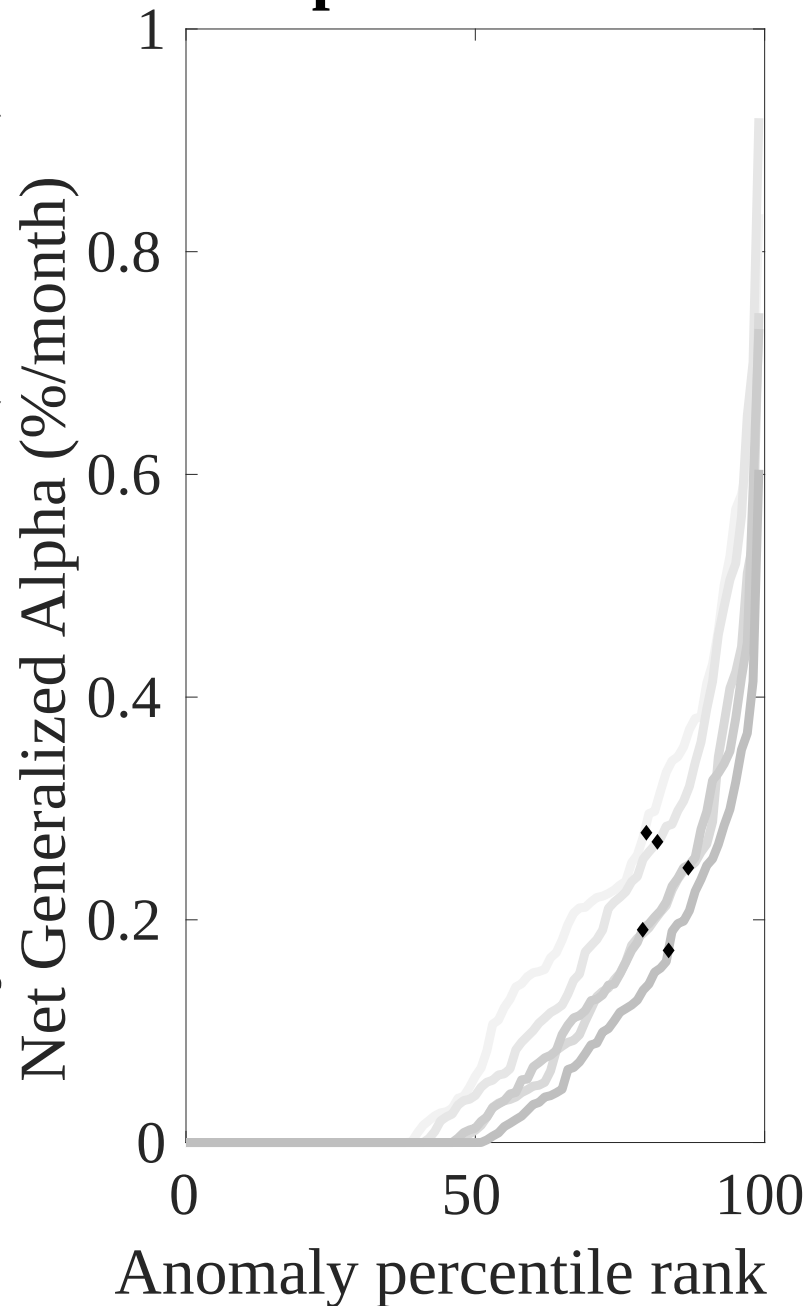
This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the IFC trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.

Gross Alpha Percentiles



Novy-Marx and Velikov (RFS, 2016)

Net Alpha Percentiles



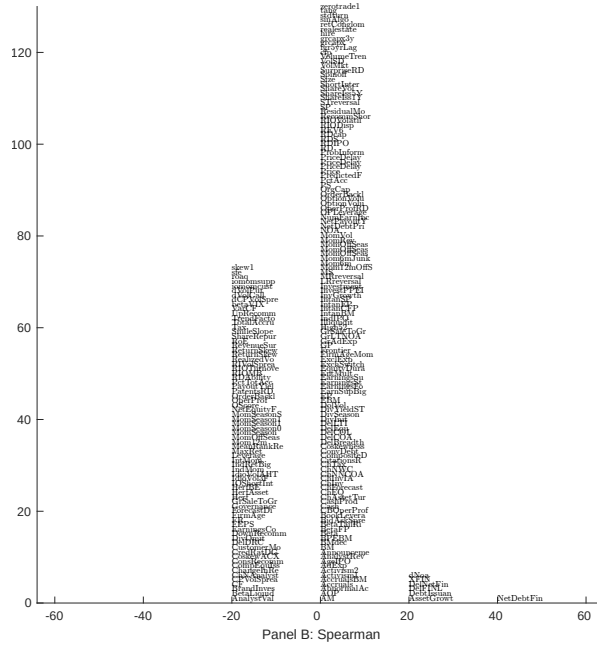
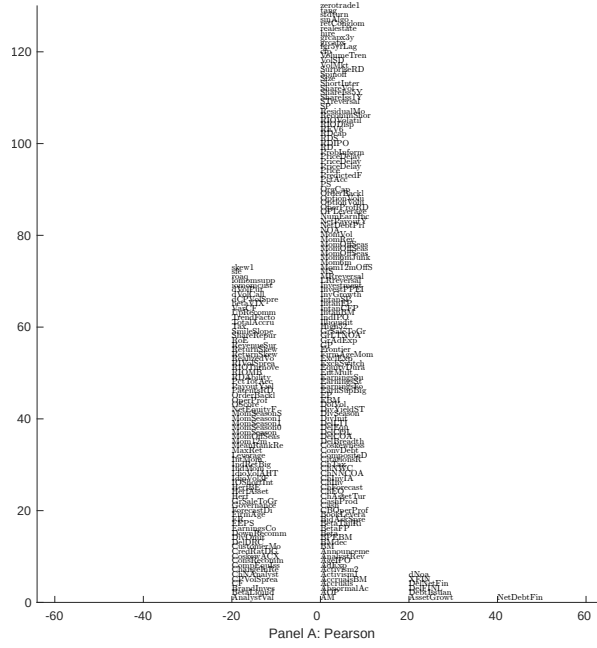


Figure 5: Distribution of correlations.
This figure plots a name histogram of correlations of 210 filtered anomaly signals with IFC. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

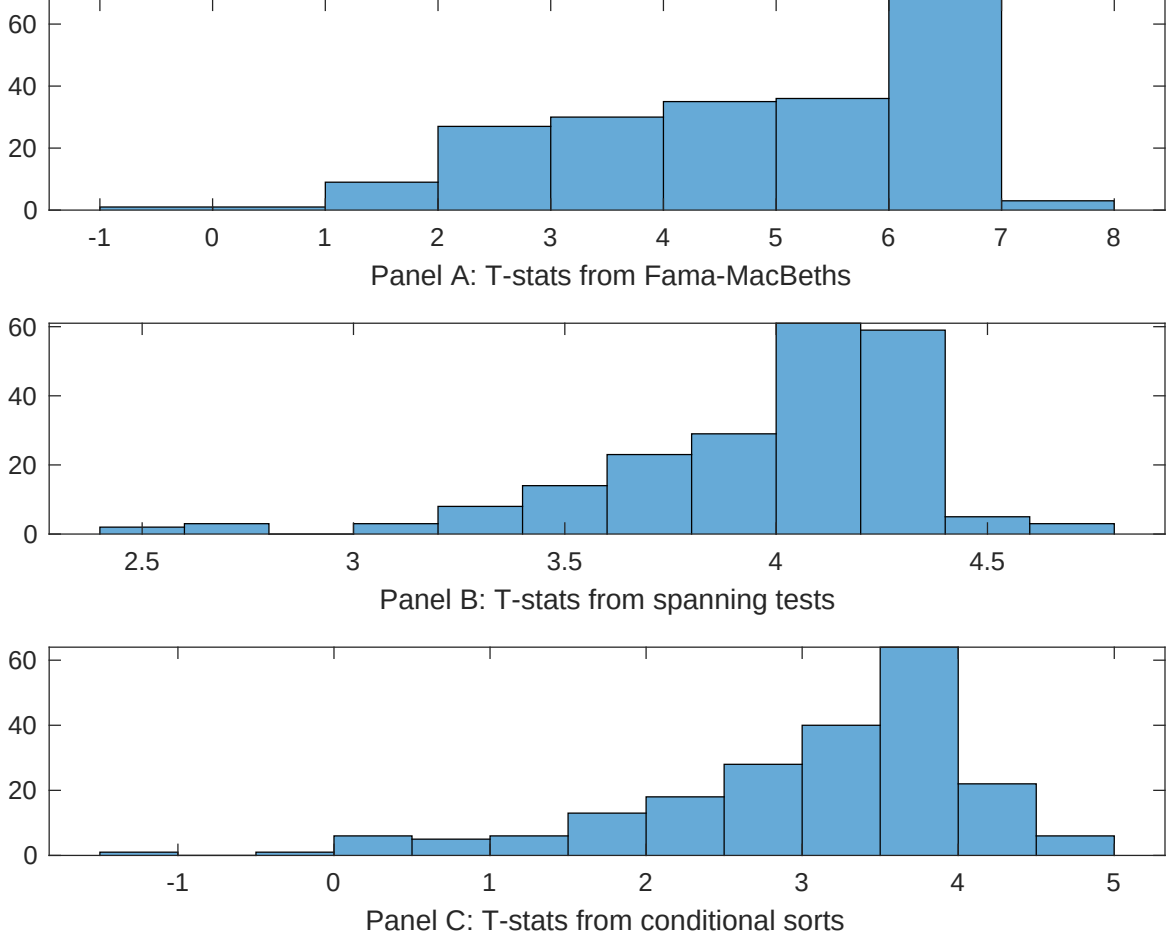


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of IFC conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{IFC} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{IFC} IFC_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{IFC,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on IFC. Stocks are finally grouped into five IFC portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted IFC trading strategies conditioned on each of the 210 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on IFC. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{IFC} IFC_{i,t} + \sum_{k=1}^s \beta_{X_k} X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X , are Change in financial liabilities, Net external financing, Net debt financing, Asset growth, change in net operating assets, Employment growth. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 197306 to 202406.

Intercept	0.13 [5.38]	0.14 [5.69]	0.13 [5.34]	0.14 [5.81]	0.14 [5.68]	0.13 [5.35]	0.15 [5.96]
IFC	0.51 [2.10]	0.80 [3.11]	0.61 [2.32]	0.54 [2.05]	0.45 [1.76]	0.13 [5.28]	0.19 [0.71]
Anomaly 1	0.17 [9.10]						-0.54 [-1.32]
Anomaly 2		0.20 [6.43]					0.80 [1.53]
Anomaly 3			0.21 [9.11]				0.60 [1.06]
Anomaly 4				0.11 [8.61]			0.49 [2.64]
Anomaly 5					0.14 [9.67]		0.73 [4.55]
Anomaly 6						0.95 [6.20]	0.12 [0.98]
# months	612	612	612	612	612	612	612
$\bar{R}^2(\%)$	0	1	0	0	0	0	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the IFC trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{IFC} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies, X , are Change in financial liabilities, Net external financing, Net debt financing, Asset growth, change in net operating assets, Employment growth. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 197306 to 202406.

Intercept	0.19 [2.86]	0.20 [2.89]	0.20 [2.93]	0.21 [3.05]	0.21 [3.01]	0.22 [3.18]	0.20 [2.90]
Anomaly 1	14.96 [3.91]						11.09 [2.10]
Anomaly 2		10.79 [3.18]					7.41 [2.02]
Anomaly 3			15.30 [4.15]				7.07 [1.40]
Anomaly 4				8.20 [2.08]			6.10 [1.41]
Anomaly 5					1.70 [0.43]		-7.28 [-1.66]
Anomaly 6						3.45 [0.94]	2.09 [0.54]
mkt	-2.57 [-1.65]	-1.43 [-0.88]	-2.88 [-1.86]	-2.91 [-1.86]	-2.91 [-1.85]	-2.82 [-1.79]	-1.59 [-0.97]
smb	-0.73 [-0.31]	4.27 [1.64]	-0.55 [-0.23]	0.21 [0.09]	0.84 [0.35]	1.04 [0.43]	0.99 [0.36]
hml	-8.39 [-2.79]	-8.18 [-2.70]	-8.76 [-2.93]	-9.77 [-3.23]	-9.51 [-3.12]	-10.04 [-3.22]	-7.62 [-2.45]
rmw	5.31 [1.73]	0.21 [0.06]	4.97 [1.62]	6.68 [2.18]	6.68 [2.17]	6.81 [2.21]	0.45 [0.12]
cma	23.78 [5.04]	21.75 [4.29]	24.82 [5.35]	18.86 [2.85]	27.52 [5.08]	25.70 [4.55]	14.55 [2.08]
umd	2.47 [1.56]	3.69 [2.38]	2.78 [1.78]	3.93 [2.52]	3.71 [2.37]	3.54 [2.24]	2.52 [1.59]
# months	612	612	612	612	612	612	612
$\bar{R}^2(\%)$	13	13	14	12	11	11	15

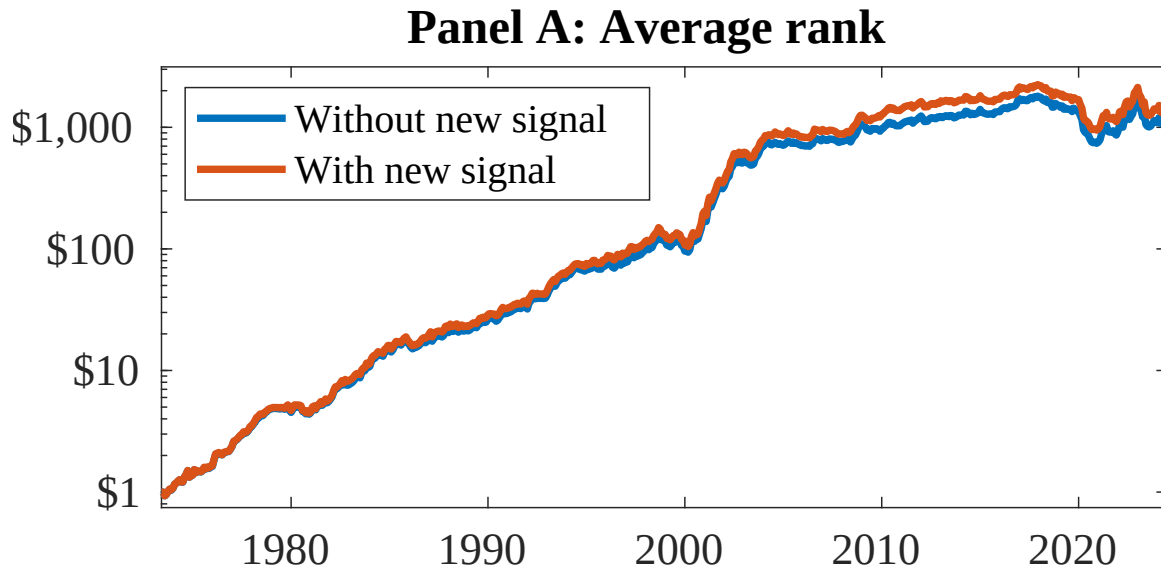


Figure 8: Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 155 anomalies. The red solid lines indicate combination trading strategies that utilize the 155 anomalies as well as IFC. Panel A shows results using "Average rank" as the combination method. See [Section 7](#) for details on the combination methods.

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